Received: January 08,, 2023 | Accepted: January 29, 2023 | Published: February 05, 2023

Crossref PEERCITE PUBLISHERS Open Access

Peercite - Journal of Surgery and Medicine

ORIGINAL RESEARCH

Exonic Variation of Asic5 Gene Uncovered by Whole-Genome Sequencing Leads to Recurrent Pregnancy Loss

Sayed AbdulAzeez [1], **Noor B. Almandil** [2], **Norah Fahad Alhur** [1]

[1] Department of Genetic Research, Institute for Research and Medical Consultations, Imam Abdulrahman Bin Faisal University, Dammam, Saudi Arabia

[2] Department of Clinical Pharmacy Research, Institute for Research and Medical Consultations, Imam Abdulrahman Bin Faisal University, Dammam, Saudi Arabia

★ *Corresponding Author : Sayed AbdulAzeez*

Email : Sayed AbdulAzeez@gmail.com

Copyright: © 2023 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https:// peercite.org).

Abstract :

Family trio next-era sequencing-primarily based version evaluation became executed to pick out the genomic motive on unexplained recurrent being pregnant loss (RPL). An own family (dead fetus and parents) from Saudi Arabia with an earlier history of 3 unexplained RPLs at the ninth week of being pregnant was included in the have a look at. Whole-genome sequencing (WGS) of a lifeless fetus and the parents became achieved to identify the pathogenic version and showed via Sanger sequencing. WGS of dead fetus identifies a singular homozygous exonic version (NM_017419.3:c.680G>T) in ASIC5 (acid-sensing ion channel subunit member of the family 5) gene; the parents are heterozygous. Newly designed fingers PCR followed by using direct sequencing confirms the presence of heterozygous in a single subject and lack of homozygous novel mutation amongst randomly decided on healthy Saudis. The second circle of relatives with heterozygous changed into showed with three unexplained RPLs. Pathogenicity analysis of R227I amino acid substitution in ASIC5 protein via molecular docking and interplay evaluation found out that the mutations are exceptionally pathogenic, decrease the steadiness of the protein, and prevent binding of amiloride, which is an activator to open the acid-sensing ion channel of ASIC5. The recognized uncommon and novel autosomal recessive mutation, c.680G>T:p.R227I (ASIC5Saudi), in two families confirm the ASIC5 gene association with RPL and can be fatal to the fetus.

Keywords : Whole genome sequencing, Molecular docking, Next generationse quencing, Recurrent pregnancy loss

Introduction :

Recurrent pregnancy loss (RPL), or recurrent miscarriage (RM) is described as three or extra sequential unpremeditated abortions earlier than 20 weeks of gestation, a situation termed “ordinary abortion” or “repeated spontaneous abortions”. RPL affects couples at propagative age round the world. The etiologies of RPL in Saudis or Arabs and different populations tend to be multifactorial.

Factors including genetic abnormalities, placental anomalies, mental trauma and annoying existence activities, and certain coagulation and immunoregulatory protein defects were stated to be associated with RPL among women in the Gulf vicinity.In a few populations, other factors were studied, including anatomical, endocrine, hormonal problems, infection, smoking and alcohol consumption, and exposure to environmental elements, and these elements may want to growth the hazard of RPL [1].

Numerous studies have suggested the connection among various reasons of recurrent miscarriage amongst Saudis and the relaxation of the populace; however, 30–50% of RPLs had been unexplained more research on RPL best can monitor the reason. The goal of the examine is to analyze the genetic foundation of an own family from Saudi Arabia with an earlier records of recurrent pregnancy loss at the ninth week of being pregnant using subsequent-era sequencing whole-genome sequencing (WGS) by means of whole evaluation of whole genome of the fetus and dad and mom accompanied through rigorous bioinformatics and confirmatory analyses. The study reports a singular homozygous exonic variation inside the ASIC5 gene in a dead fetus, whilst the mother and father are heterozygous [2].

Materials and methods:

Ethics and study subjects:

An own family with a beyond records of three miscarriages has been blanketed within the observe with a written consent from the moms and dads in the course of the fourth being pregnant, the mother experienced a comparable type of miscarriage on the ninth week of pregnancy. Tissue (separated cautiously from maternal tissue to keep away from contamination) samples and blood samples were collected from the fetus (proband) and parents, respectively. Miscarriage sample was gathered in an RNAprotect mobile Reagent [3]. The DNAs of the samples were remoted, and the maximum widespread genetic ailment, hemoglobinopathies, had been screened the usage of the Sanger sequencing. Genes (purposeful variations and deletions in HBB, HBA1, HBA2, ATRX, and HBD) associated with the most accepted mutations have been found to be every day. Therefore, the WGS turned into carried out for the miscarriage tissue, mother, and father genomes.

Entire-Genome Sequencing and Trio evaluation:

The trio evaluation has been finished the usage of the satisfactory practice GATK pipeline. The program Fastx (http://hannonlab.cshl.edu/fastx_toolkit) became used to clear out low-excellent reads. Then the reads were aligned to the reference human genome (hg19) the usage of this system BWA. The GATK haplotype caller changed into used to call the variations [4].

The resulting variants have been then annotated using in-residence evolved workflow which include the subsequent 3 sets of records resources:

1. Public databases: those were accumulated from the Annovar packages, and that they include the simple positional information approximately genes and associated proteins. In addition, they include data from the dbSNP database, the 1000 Genome database, ExAC, and gnomAD databases [5]. Annovar additionally includes predictions of the purposeful effect of the variants from the equipment Polyphan, Sift, CADD, and Meta SVM. Similarly, to Annovar, we used the clinvar and OMIM databases to annotate the editions and genes with up-to-information medical information.

2. In-house databases: We annotated the versions the usage of the Saudi Human Genome application variant DB to test for variant frequency within the Saudi population.

3. Commercial databases: We used the HGMD database to annotate the variant with clinical information.

After variation annotation, we ran filters in step with the ACMG (American college of scientific Genetics and Genomics) suggestions. We excluded variations which might be intergenic, synonymous, acting greater than 5% in populace databases, or now not detrimental (as predicted by means of CADD, Polyphen, SIFT, and MetaSVM).

We also ran greater trio evaluation to filter out the variations consistent with the autosomal recessive, de novo, compound heterozygous, and x-connected. After applying these filters, the remaining versions were tested manually to fit the annotated scientific records to the fetus phenotype [6].

Sanger Sequencing Validation:

Whole-genome end result turned into showed the use of Sanger sequencing. The presence of the homozygous NM_017419.3:c.680G>T in the proband and heterozygous within the mother and father had been confirmed the usage of Sanger sequencing. Surprisingly precise primers (ASIC5F: 5′-CAGATAAAAACATGTTTCCATACATCTTCAG-3′ and ASIC5R: 5′- TTGTGGCATGAACATTCCCTGGA-3′) were designed, and the selected region of the gene was amplified .

[PCR recipe: MOLEQULE-ON absolute master mix 12.5 µl, ASIC5F 1 µl (10 nM), ASIC5R 1 µl (10 nM), DNA Template 25 ng, and Dis H2O to 25 µl; temperature profile: 95◦C for 10 min; 35 cycles of 95◦C/60 s, 60◦C/60 s, 72◦C/60 s; and 72 ◦ C Sequencing Kit (Thermo FisherScientific, Inc., Waltham, MA, USA). Amplified PCR product (691 bp) of the ASIC5 gene region was purified and sequenced using Genetic Analyzer 3500 (Thermo Fisher Scientific, Inc.) at the Department of Genetic Research, Institute for Research and Medical Consultations, Imam Abdulrahman Bin Faisal University (Dammam, Saudi Arabia). Sequences were analyzed using mutation surveyor software (Softgenetics, US) and DNA sequencing analysis software v.5.3 (Applied Biosystem; Thermo Fisher Scientific, Inc.) [7].

Amplification Refractory Mutation gadget-Polymerase Chain reaction-primarily based variation Screening and Sanger Sequencing Validation:

The amplification refractory mutation system-polymerase chain reaction (ARMS-PCR) was designed (primers will be available on request) to screen the presence of NM_017419.3:c.680G>T among healthy Saudis (n = 200). The subjects positive for the presence ofNM_017419.3:c.680G>T was confirmed through Sanger sequencing using primers (ASIC5F and ASIC5R). This is also to confirm the absence of the homozygous NM_017419.3:c.680G>T in the healthy Saudi subjects randomly selected [8, 9].

Homology Protein Modeling and practical Annotations:

The homology modeling of wild (p.R227) and mutant (p.R227I) ASIC5 protein became finished the use of Swiss version server, tested the usage of PROCHECK. The structural functional annotations were completed the use of SAS-series server, ProFunc and PDBsum. Mutant systems have been generated using Swiss-PDB Viewer and PyMol. Energy minimization for the wild and mutants became estimated the use of GROMACS. Evolutionary conservation and practical component analysis of the R227 residue inside the wild-type protein become done the usage of the ConSurf [10]. PROVEAN and that I-Mutant were used for studying the effect at the organic characteristic of a protein because of an amino acid substitution R227I.

AutoDock Vina become used for molecular docking of the ligand with wild type and mutant ASIC5 protein, and the molecular visualization become finished in PyMol and LigPlot.

Results:

Whole-Genome Sequencing and Trio evaluation:

His family with a history of three unexplained miscarriages was protected in the study. The couple is consanguineous but now not first-degree spouse and children.

There were no records of genetic and chronic diseases inside the couple. The family turned into diagnosed with a similar kind of unknown spontaneous abortion at the ninth week of pregnancy. The mom changed into 30 years on the time of the fourth unexplained spontaneous miscarriage; the daddy was 34.

The preceding three unexplained miscarriages and the fourth have been additionally of comparable gestation. At this gestation, the gender of the proband cannot be decided even after miscarriage. The mom is without uterine or cervical abnormalities [11].

If you want to perceive the motive of the recurrent spontaneous abortion, WGS became finished for the mom, father, and proband.

The WGS of the trio (proband and dad and mom) samples has found out an inheritance of NM_017419.3:c.680G>T mutation within the ASIC5 gene from the dad and mom (determine 1A and Supplementary table 1).

Numerous heterozygous mutations located inside the proband are indexed within the Supplementary material, which have been inherited either from the dad or mum.

The WGS end result of NM_017419.3:c.680G>T variation in exon four of the ASIC5 gene has been confirmed via the Sanger sequencing. the father and the mother have been located to be vendors (heterozygous) of the c.680G>T:p.R227I at the ASIC5 gene, while the proband turned into homozygous to c.680G>T:p.R227I (GenBank: MN251164; ClinVar: SCV000930628; SNP id: rs1248841709) (figure 1).

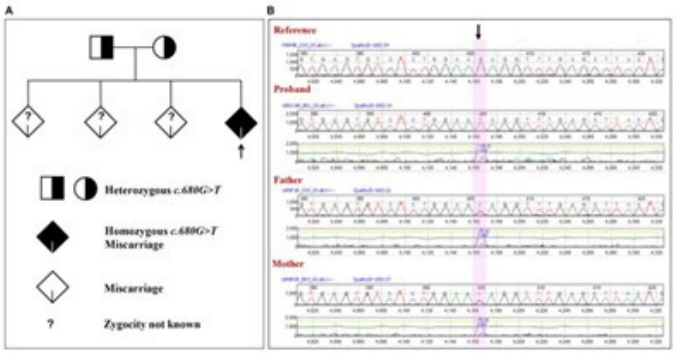

Figure 1: Novel mutation in the ASIC5 gene (NM_017419.3:c.680G>T) in the family.

Amplification Refractory Mutation system-Polymerase Chain response-based totally variation Screening and Sanger Sequencing Validation:

If you want to affirm the absence of the homozygous NM_017419.three:c.680G>T many of the residing population, a total of 200 healthful Saudis have been decided on randomly and checked for the mutation at the c.680 position within the ASIC5 gene using arms-PCR accompanied by using Sanger sequencing.

The consequences of the fingers-PCR and direct sequencing of 200 healthful Saudis within the c.680 function in the ASIC5 gene found out the absence of homozygous NM_017419.3:c.680G>T.

Furthermore, this mutation is novel to the SHGP (Saudi Human Genome application) database (approximately 9, 500 instances).

This indicates that the located mutation NM_017419.3:c.680G>T is rare, and their absence of a homozygous nation within the healthy Saudis is validated.

Furthermore, a lady concern became determined with a heterozygous NM_017419.3:c.680G>T in the ASIC5 gene.

The lady problem with heterozygous mutations is an unmarried daughter, and her mother skilled the unexplained RPL similar with the earlier circle of relatives in the 9th week of pregnancy consecutively three times [12].

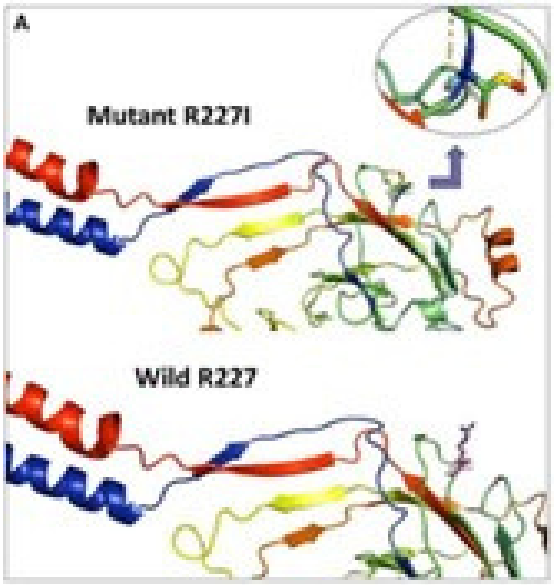

Figure 2:Structural models of the wild (R227) and mutated (R227I) ASIC5 proteins.

Molecular Docking and interaction evaluation:

The anticipated structure of the wild ASIC5 on the Ramachandran plot confirmed φ/9 angles of 83.1% residues within the most favored regions, 15.4% in the additional allowed areas, 1.1% within the generously allowed areas, and 0.3% inside the disallowed areas. The overall residue span of the secondary shape consists of 23.0% residues involved within the formation of the strands, 23.0% residues in alpha helices, 2.6% residues in 3-10 helices, and 51.5% residues in different structural moieties.Analysis of secondary structure in ProFunc confirmed the presence of three βsheets, four β-hairpins, 1 psi loop, three β-bulges, 14 strands, 14 helices, 5 helix-helix interactions, 34 β-turns, and 9 γ-turns (figure 3) [13].

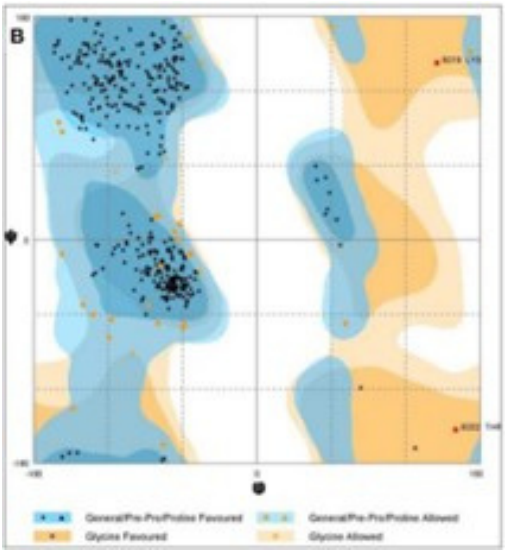

Figure 3: Filled bars show AgNP-treated mice

Homology modeling of the mutant structure (R227I) of the ASIC5 showed deviations from the wild type; the mutant structure on the Ramachandran plot showed φ/nine angles of 84.3% residues within the most preferred areas, 14.3% within the extra allowed areas, 1.1% in the generously allowed areas, and 0.3% in the disallowed regions (figure 4).

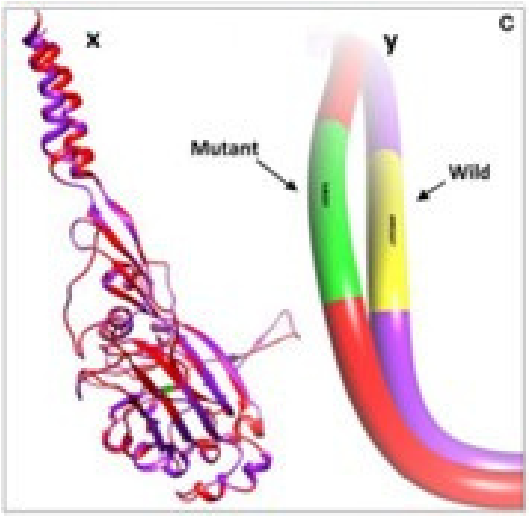

Figure 4: Superimposed structures of the wild (R227) and mutated (R227I) ASIC5 proteins.

The total residue span of the secondary mutant shape consisting of twenty-two.7% residues regarding the formation of the strands, 23.7% residues in alpha helices, 1.8% residues in 3-10 helices, and 51.8% residues in other structural moieties. analysis of the secondary structure of the mutant in ProFunc showed the presence of 3 β-sheets, 4 β-hairpins, 1 psi loop, 2 β-bulges, 14 strands, 13 helices, 5 helix-helix interactions, 42 β-turns, and 8 γ-turns (figure 5) [14].

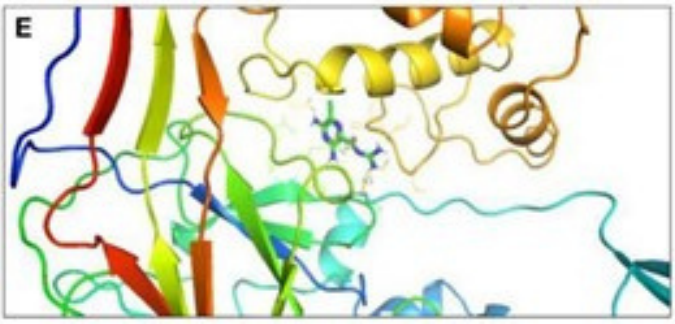

Figure 5: 3D amiloride with surrounding amino acids of ASIC5 protein.

Discussion:

Studies on tissues of miscarriage specimens from girls with RPL observed the chromosomal aberrations from 29 to 46% of miscarriage tissues, even as majority of the RPL may be due to opportunity mechanisms or aside from chromosomal aberrations. The existing commentary indicates that coding editions in ASIC5 gene may be one a few of the opportunity mechanisms for RPL. The position of the acid-sensing ion channel subunit member of the family five (ASIC5) or ACCN5 or bile acid-touchy ion channel gene in people, in widespread, and the development of the fetus, specially, is scanty. Very constrained research is to be had on the gene ASIC5 and associated expression. This gene, ASIC5, became reported to be expressed inside the amniotic fluid, fetal intestine, brain, liver, coronary heart, ovary, and testis. ASIC5 is overexpressed in the fetal gut and plasma. ASIC5 became discovered to a key participant within the body structure of unipolar brush cells of the vestibulocerebellum. The entire capabilities of the ASIC5 gene and its product are but to be diagnosed. Animal research on the autosomal recessive mouse mutant of the gene encoding the L-kind calcium channel found out that the homozygous mutant animals die at birth; however, the heterozygous for the mutant isn't always distinguishable from that of wild animals. The look at resembles the existing commentary of the heterozygous mutant of the wholesome parents, whilst demise of the fetus with a homozygous mutant in the gene belongs to the amiloride-sensitive Na+ channel.

The R227I earlier reports said that in 39% of the Saudi women who had RPL, the origin of the affected person in the look at turned into unexplained or had no identifiable motive. Diverse motives including genetic factors had been stated for recurrent being pregnant loss among Saudi girls. Consanguineous marriages also are extensively ($p = 0.046$) impacting. Genome-huge affiliation takes a look at (GAWS) found out the affiliation of ASIC5 ($p = 0.0029$) and level of manganese. Moreover, the extent of manganese within the placental tissue of Saudi girls with recurrent pregnancy loss changed into substantially ($p < 0.0001$) decreased. This suggests that the identified mutation in ASIC5 would possibly have played a role inside the degree of manganese inside the gift girls. A recent examine on the analysis markers of glioblastoma discovered the expression of ASIC5 as associated diagnosis markers. ASIC5 became determined to be activated within the ethanol-(a hundred mM)-exposed neonatal rat cardiomyocytes at the side of other six molecules (CYP2A6, PRL, CHRNA4, CNR1, CRH, and SLC40A1).

Low (in 50%) ASIC5 protein expression in melanoma have been determined with <4% mutation rates.

Preparing the mutated animal model to study the impact of the mutant on the fetal development is not available in our laboratory, which is a limitation of the study. Hence, the region with the mutation, c.680G>T in the ASIC5 gene, became screened the usage of palms-PCR followed by sequencing using designed primers to discover the presence of c.680G>T in randomly selected Saudis inside the have a look at vicinity, which confirms the absence of the homozygous NM_017419.3:c.680G>T and the presence of heterozygous NM_017419.3:c.680G>T in a lady subject and her mother with RPL. The take a look at confirms the impact of the association of the novel exonic mutation with RPL. However, nationwide studies are mandatory to perceive the superiority of this uncommon mutation and mutations on this gene amongst unexplained miscarriages cases and their effect on the recurrent being pregnant loss and fetal improvement. This may reveal the position of ASIC5. The protein-protein interaction analysis of ASIC5 protein, with the protein found with the mutation in the proband the use of STRING, discovered loss of interaction. But, the analysis the use of STRING cannot monitor any particular impact of mutated protein-protein interactions due to unique amino acid adjustments.

Primarily based on the earlier reports on the member of the DEG/ENaC (degenerin/epithelial sodium channel) protein family and the cutting-edge observations, it could be concluded that the R227I amino acid substitution inside the ASIC5 is enormously deleterious; the mutant ASIC5 showed reduced stability and of the protein and forestalls the binding of amiloride, a robust inhibitor of acid-sensing ion channel proteins.

Conclusion:

The located novel ASIC5 gene-coding variation (ASIC5Saudi) in families verify the ASIC5 affiliation with the consequences of RPL. hence, this mutation is pathogenic, which may additionally motive severe infection to the fetus and motive fetal mortality. The molecular mechanism at the back of the dying of the fetus with regards to the homozygous NM_017419.3:c.680G>T at exon 4 (ASIC5Saudi) in the ASIC5 gene should be studied in detail. Early prenatal prognosis of pathogenic variation like ASIC5Saudi can offer a choice for the determine to decide being pregnant termination inside the allowed time amongst high-consanguinity population.

Acknowledgment:

The authors thank the Dean, Institute for Research and Medical Consultations (IRMC), Imam Abdulrahman Bin Faisal University, Dammam, Saudi Arabia, for her continuous support and encouragement. We thank the Saudi Human Genome Program, King Abdulaziz City for Science and Technology team, for their support. The authors thank Mr. Ranilo M. Tumbaga, Mr. Horace T. Pacifico, and Mrs. Jee Entusiasamo Aquino for their assistance.

Conflict of interest:

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References:

1.Turki RF, Assidi M, Banni HA et al. (2016) Associations of recurrent miscarriages with chromosomal abnormalities, thrombophilia allelic polymorphisms and/or consanguinity in Saudi Arabia. BMC Med Gen. 17:69.

2.Al-Ghamdi AA, Makhashen SF (2016) Etiology of recurrent pregnancy loss in Saudi females. Saudi J Med Med Sci. 4:187.

3.Tan Y, Yin X, Zhang S et al. (2014) Clinical outcome of preimplantation genetic diagnosis and screening using next generation sequencing. Gigascience. 3:30.

4.Ghneim HK, Al-Sheikh YA, Alshebly MM et al. (2016) Superoxide dismutase activity and gene expression levels in Saudi women with recurrent miscarriage. Mol Med Rep. 13:2606-2612.

5.Gowri V, Udayakumar AM, Bsiso W et al. (2011) Recurrent early pregnancy loss and consanguinity in Omani couples. Acta obstetricia et gynecologica Scandinavica. 90:1167-1169.

6.McKenna A, Hanna M, Banks E et al. (2010) The genome analysis toolkit: a MapReduce framework for analyzing next-generation DNA sequencing data. Genome Res. 20:1297-1303.

7. Li H, Durbin R (2010) Fast and accurate long-read alignment with Burrows-Wheeler transform. Bioinformatics. 26:589-595.

8.Abouelhoda M, Sobahy T, El-Kalioby M et al (2016) Clinical genomics can facilitate countrywide estimation of autosomal recessive disease burden. Genet Med. 18:1244.

9.Lindahl E, Azuara C, Koehl P et al. (2006) NOMAD-Ref: visualization, deformation and refinement of macromolecular structures based on all-atom normal mode analysis. Nucleic Acids Res. 34:52-56.

10.Choi Y, Chan AP (2015) PROVEAN web server: a tool to predict the functional effect of amino acid substitutions and indels. Bioinformatics. 31:2745-2747.

11.Trott O, Olson AJ (2010) AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. J Comput Chem. 31:455-461.

Citation: AbdulAzeez S, et al. (2023) Exonic Variation of ASIC5 Gene Uncovered by Whole-Genome Sequencing Leads to Recurrent Pregnancy Loss. Peercite J Surg Med. 1: pages.

12.Li TC, Makris M, Tomsu M et al. (2002) Recurrent miscarriage: aetiology, management and prognosis. Hum Reprod Update. 8:463-481.

13.Boiko NY, Kreko-Pierce T, Pugh J et al. (2019) Ataxia in the ASIC5 knockout mouse associated with decreased excitability of vestibulocerebellum unipolar brush cells. FASEB J. 33:824-813.

14. Mashimo K, Ohno Y (2020) Effect of ethanol on gene expression of beating neonatal rat cardiomyocytes-further research with ingenuity pathway analysis software. J Nippon Med School. 88:209-219.